From DNA to Protein

Biology 2e chapter 15 reading connection | Teacher guide / classroom preparation

Learning target

What evidence links a DNA template and translation to production of a fluorescent protein?

Timing and materials

Kit-dependent: a setup class plus observations after incubation.

Educational cell-free expression kit with reporter DNA, negative control and an approved translation-blocking condition, appropriate micropipette and tips, tubes and racks, kit-compatible fluorescence viewer and prescribed incubation/storage equipment.

Classroom route

1. Select a teaching kit that supports these comparisons; verify its controls, storage and viewer requirements before ordering. Follow the official kit protocol for quantities and preparation rather than substituting model settings.

2. Prepare three independent reactions for each available condition, holding volume and incubation conditions constant. If the kit does not supply an approved inhibitor condition, run the two supported conditions and record the difference from the virtual plan.

3. Observe under consistent lighting and camera/viewer settings at the kit-specified intervals. Use instrument readings if available; otherwise record a defined ordinal brightness score and label it as such.

Pilot the physical procedure with your own equipment. Follow school handling and disposal rules and the purchased kit instructions. The virtual numerical settings are not an equipment prescription.

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Hybrid or substitute assignment

Assign all virtual, all physical, or specified samples in each mode. The classroom entry button is available during collection. Students complete the same prediction and evidence questions. Record classroom time series on the worksheet; the online classroom form stores endpoint measurements and notes. Keep sources and measurement scales separate.

Guided virtual sequence

Students prepare each sample, inspect the apparatus or classify specimens, then explicitly collect data. All assigned trials are required before exploration. Reload pauses the clock and restores saved work. Downloads retain source labels; no automatic grade is submitted.

Expected reasoning

The reporter-DNA condition develops fluorescence; no-DNA and translation-blocked conditions remain near background in this model. Reporter maturation delays visibility. This is not direct mRNA measurement, and fluorescent signal depends on more than transcription.

Feedback and scoring

Criterion	Points	Evidence
Prediction and controls	0-2	Testable claim; comparable samples and controlled variables.
Observations	0-2	Complete, labeled records with units and source.
Interpretation	0-2	Correct calculations or categorical comparisons; claim tied to evidence.
Limitations	0-2	Specific uncertainty and a useful follow-up investigation.

Score reasoning and data quality, not agreement of physical results with the simulation. Classification feedback is a practice aid and does not overwrite student scores.

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Model limitations

Generic single-reporter model in arbitrary units; not a reproduction or calibration of BioBits. Transcription and translation are not independently measured. Classroom ordinal brightness and numerical virtual fluorescence must not be pooled.

Troubleshooting

Confirm the preparation steps are finished before starting or collecting. Counting activities require every specimen plus a review; cell measurements require a focused 40x view; gels require staining and three size estimates; fluorescence labs require the viewer. Use the visible Collect data and continue button to advance.

For physical work, preserve unexpected observations, check the control and instrument setup, then repeat with fresh material where appropriate. Never replace measured results with the virtual answer.

Evidence and optional classroom resources

Optional cell-free classroom kit

https://www.minipcr.com/product/biobits-central-dogma/

External resources remain on their publishers' websites. This packet contains original text and tables, not supplier worksheets. Chapter numbers are reading connections. No supplier or textbook publisher endorsement is implied.

Teacher pilot checklist

Confirm availability and preparation time; test controls; check units and observation windows; verify handling/disposal; choose class grouping and accessible alternatives; keep notes for revision.